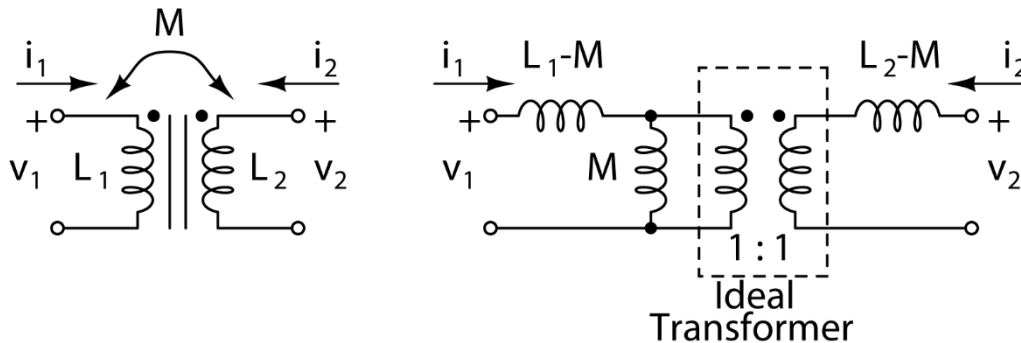


EE202 LAB #5

COUPLED INDUCTORS and MUTUAL INDUCTANCE

Preliminary Work:

Consider two inductors, L_1 and L_2 , coupled to each other with a mutual inductance of M . The dots in the left schematic show the sense of coupling. The right figure shows the equivalent circuit of the coupled inductors containing a 1:1 ideal transformer.

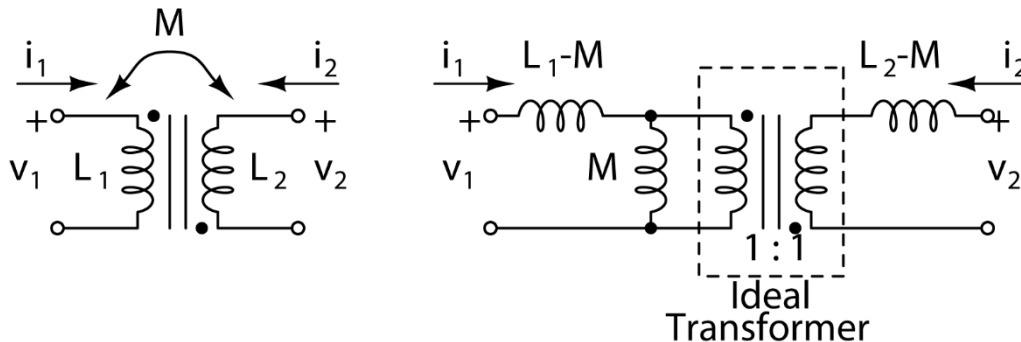


We have the following phasor equations that are valid for both circuits:

$$V_1 = j\omega L_1 I_1 + j\omega M I_2 \quad \text{and} \quad V_2 = j\omega L_2 I_2 + j\omega M I_1$$

If $I_2 = 0$ (L_2 is open-circuited), $V_1 = j\omega (L_1 - M) I_1 + j\omega M I_1 = j\omega L_1 I_1$ (we only have L_1).

The sense of coupling can be reversed when one of the dot's position is changed as shown below.



If the sense of coupling is reversed we have

$$V_1 = j\omega L_1 I_1 - j\omega M I_2 \quad \text{and} \quad V_2 = j\omega L_2 I_2 - j\omega M I_1$$

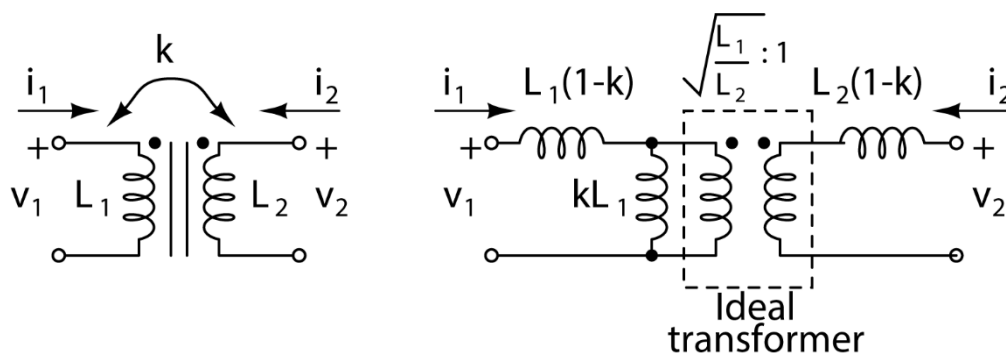
1. Using the equivalent circuit, find the total inductance in terms of L_1 , L_2 , and M if the coupled inductors L_1 and L_2 are connected in series.
2. Find the total inductance if the direction of L_2 is reversed in the series connection.
3. One can use the total inductance difference between the two series connections to measure the mutual inductance. Explain how.

An alternative method of representing the mutual inductance between two coils is the coupling coefficient, k . LTSpice utilizes this method. The coupling coefficient k and mutual inductance M are related to each other by the equation

$$M = k (L_1 L_2)^{1/2}$$

where $0 \leq k \leq 1$. k shows the percentage of magnetic flux lines crossing both L_1 and L_2 . If $k=0$, the two inductors are not coupled. If $k = 1$, the coils are perfectly coupled, which never happens in real life.

Since it is possible to have $M=k(L_1 L_2)^{1/2} > L_2$ (or L_1), the equivalent circuits above using the mutual inductance M may contain negative inductance values (e.g., $L_2 - M < 0$). An alternative equivalent circuit that does not have any negative valued inductor is given below (Unlike the first equivalent circuit, the turns-ratio of the ideal transformer is *not* 1:1):



4. Using this equivalent circuit, find the open-circuit voltage across the second inductor, v_2 , in terms of L_1 , L_2 , and k if a voltage source (v_1) of unit amplitude is applied to the first. This is the effective turns ratio of the transformer formed by two coupled inductors.
Hint: $L_2(1-k)$ inductor has no voltage drop across it since $i_2=0$. kL_1 and $(1-k)L_1$ form a voltage divider circuit.

5. Two coupled inductors in LTSpice: First, insert two inductors L_1 and L_2 . Choose their values in the range $10\mu\text{H}$ to $150\mu\text{H}$. Choose a coupling coefficient, k , with $|k| < 1$. Then, insert the spice directive: "K L1 L2 xx" where xx is the coupling coefficient k . As soon as you insert this directive, the inductors will have dots associated with the sense of coupling.

Verify your finding in part 1 by connecting the coupled inductors in series. The total inductance can be found by applying a current through the inductors and measuring the voltage across the inductors. Perform an AC analysis in the frequency range 50KHz to 100KHz by setting AC amplitude of the current source to 1. Plot the voltage across the inductors. Change the vertical scale to linear by right-clicking on the vertical axis. The inductor value can be read directly if the total inductor voltage is divided by $(2\pi \cdot \text{freq})$. Note that the vertical axis unit will be volts/Hz rather than Henries.

6. Verify your finding in part 2 by reversing the direction of one of the coils in the series connection.
7. Find the mutual inductance (see part 3) by using the results of parts 5 and 6. Compare the mutual inductance found by the value $k(L_1 L_2)^{1/2}$.

- Assume that the inductors have a quality factor of $Q=60$ at the operating frequency (a chosen frequency of 50KHz to 100KHz). The equivalent series resistance, r , of the inductor can be found from

$$r = \omega L / Q$$

where $\omega = 2\pi f$ is the operation frequency.

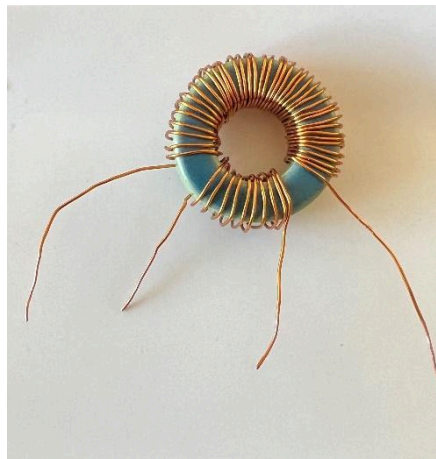
In LTSpice, insert resistors in series with the inductors with values calculated from the formula above. Apply an AC amplitude=1 voltage source. Perform an AC analysis between frequencies 99KHz and 101KHz. Determine V_2 at 100KHz on the linear scale to find the effective turns ratio and verify your result for part 4. Find the phase shift (degrees) in V_2 at the same frequency.

- Repeat step 8 when a 10Ω resistor is across L_2 .

Experimental Work:

Coupled inductors

- This lab utilizes the core CA106-52A. It has $A_L = 106 \text{ nH/T}^2$. Wind two coils on this core. **Choose a value for N_2 in between 10 and 40. N_1 has already been assigned to you.** Remove the enamel on the wires for good electrical connections. You may want to solder-coat the wire ends. An example winding with $N_1=40$ and $N_2=10$ is shown below.



N_1	
N_2	

- Measure the inductance of L_1 and L_2 and the quality factor Q at $f=100\text{KHz}$ using the RLC meter. If Q is smaller than 30, you probably have poor electrical connection (unremoved enamel) at the wire ends.

Measure the total inductance when the inductors are connected in series. Reverse the direction of the series connection and measure again.

Calculate the equivalent series resistance of each inductor from $r = 2\pi f L / Q$. Compare the measured inductance values L_1 and L_2 with those calculated from $A_L N_1^2$ and $A_L N_2^2$.

	Measured (μH)	Measured Q	r (Ω)	$A_L N^2$ (μH)
L_1				
L_2				

	Measured (μH)	Measured Q
L_1 and L_2 in series		
L_1 and reversed L_2 in series		

Calculate the mutual inductance, M , using the results of Preliminary part 3. Calculate the coupling coefficient, k , of the two inductors from the formula $k=M/(L_1 L_2)^{1/2}$.

M (μH)	
k	

- Connect the signal generator to L_1 . Adjust the frequency of the signal generator to $f=100\text{KHz}$ (with a period of $10\mu\text{s}$). Set the amplitude to 5Vpp . Use EXT SYNC cable between the signal generator and the oscilloscope. Since the signal generator has a series resistance of 50Ω , the input voltage across the coil should also be measured. Connect the oscilloscope probe across L_1 . Adjust the oscilloscope's horizontal scale so as to see the rising zero crossing of the sine wave zoomed in. Adjust the horizontal position so that the zero crossing is in the middle of the screen. Store the V_1 waveform in the oscilloscope memory. Now, measure the voltage across L_2 . Display the stored waveform to find the time delay between the input and output. To measure the phase difference in degrees, find the time difference between the zero crossings of the stored V_1 waveform and the measured V_2 waveform, then multiply the result by $360^\circ/(10\mu\text{s})$. You may need to change the vertical scale of V_2 waveform for a more accurate phase measurement. Ensure that the zero-voltage level in the stored waveform and the measured waveform are the same.

f (KHz)	
Measured V_1 (volts p-p)	
Measured V_2 (volts p-p)	
Effective turns ratio (V_2/V_1)	
Phase difference (degrees)	

4. Repeat the measurement step 3 when a 10Ω resistor is connected across L_2 .

f (KHz)	
Measured V_1 (volts p-p)	
Measured V_2 (volts p-p)	
Effective turns ratio (V_2/V_1)	
Phase difference (degrees)	

Show your measurement results to the assistant to get a check.

5. Using LTspice, perform an AC analysis from 99KHz to 101KHz to find the open-circuit voltage. Insert the measured inductor values and calculated series resistors. Use the measured k value in the simulation. Determine the value of V_2 , across L_2 when a voltage source of unit magnitude (AC amplitude) V_1 is applied to L_1 . The phase difference can be read directly at 100KHz. Compare with the measured results.

f (KHz)	
Applied V_1 (volt)	
Simulated V_2 (volt)	
Effective turns ratio (V_2/V_1)	
Phase difference (degrees)	

6. Repeat step 5 when a 10Ω resistor is connected across L_2 . Compare with the measured results.

f (KHz)	
Applied V_1 (volt)	
Simulated V_2 (volt)	
Effective turns ratio (V_2/V_1)	
Phase difference (degrees)	